Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: J&M Dairy

Physical address of dairy:

Number and Street	City	County	Zip Code
11910 Bruceville RD	Elk Grove	Sacramento	95757

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 04/13/1992

Regional Water Quality Control Board Basin Plan designation: Sacramento River Basin

County Assessor Parcel Number(s) for dairy facility:

X132-X240-X028-XXXX

B. OPERATORS

Duarte, Todd Anthony

Operator name: Duarte, Todd Anthony

Telephone no.: (916) 683-8056 Landline, (916) 952-8355 Cellular

Mailing Address Number and Street	City	State	Zip Code
11844 Bruceville RD	Elk Grove	CA	95757

C. OWNERS

Duarte, John Fernando

Legal owner name: Duarte, John Fernando

Telephone no.: (916) 684-2270 Landline, Cellular

Mailing Address Number and Street	City	State	Zip Code
11910 Bruceville RD	Elk Grove	CA	95757

Duarte, Margaret Mary

Legal owner name: Duarte, Margaret Mary

Telephone no.: (916) 684-2270 Landline, Cellular

Mailing Address Number and Street	City	State	Zip Code
11910 Bruceville RD	Elk Grove	CA	95757

This owner is responsible for paying permit fees.

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	65	0	0	0
Number under roof	300	60	0	0	0	0
Maximum number	300	60	65	0	0	0
Average number	300	60	65	0	0	0
Avg live weight (lbs)	1,400	1,500	1,150	0		

Predominant milk cow breed: Holstein

Average milk production: 61 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 8,976.83 *tons per reporting period*

Total nitrogen from manure: 113,589.49 *lbs per reporting period*

After ammonia losses (30% loss applied): 79,512.64 *lbs per reporting period*

Total phosphorus from manure: 18,760.86 *lbs per reporting period*

Total potassium from manure: 55,678.87 *lbs per reporting period*

Total salt from manure: 155,052.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 7,965,000 *gallons*

Total nitrogen generated: 29,063.38 *lbs*

Total phosphorus generated: 7,858.20 *lbs*

Total potassium generated: 88,966.75 *lbs*

Total salt generated: 434,925.54 *lbs*

	7,965,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	7,965,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
J & M Ag Well	Ground water
J & M Dom. #1	Ground water
J & M Dom. #2	Ground water
Wagemann Dom.	Ground water
Wagemann Large Ag	Ground water

Source Description	Type
Wagemann Small Ag	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

Date	Material type	Quantity	Reporting basis	Moisture (%)	Density (lbs/cu ft)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/31/2023	Corral solids	150.00 *ton*	Dry-weight	31.6		14,200.00	3,830.00	15,400.00		70.70
10/31/2023	Corral solids	350.00 *ton*	Dry-weight	22.2		18,300.00	5,400.00	9,010.00		0.00

No liquid nutrient exports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	12,880.02	3,726.76	8,066.93	145,076.40
Process wastewater	0.00	0.00	0.00	0.00
Total exports for all materials	12,880.02	3,726.76	8,066.93	145,076.40

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
J&M North	20	20	2	both	X132-X240-X026-XXXX X132-X240-X028-XXXX
J&M South	38	38	2	both	X132-X240-X026-XXXX X132-X240-X028-XXXX
Wagemann East	30	30	2	both	X132-X240-X048-XXXX
Wagemann West	40	40	2	both	X132-X240-X048-XXXX
Totals for areas that were used for application	128	128	8		
Totals for areas that were not used for application					
Land application area totals	128	128	8		

B. CROPS AND HARVESTS

J&M North

Field name: J&M North

10/29/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 20 Plant date: 10/29/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/02/2023	375.00 *ton*	Dry-weight		65.4	16,600.00	2,550.00	21,700.00		14.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	184.00	28.80	192.00	0.00
Total actual harvest content	18.75	215.39	33.09	281.56	1,829.48

J&M North

06/08/2023: Corn, silage

Crop: Corn, silage Acres planted: 20 Plant date: 06/08/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/23/2023	587.50 *ton*	Dry-weight		74.8	13,900.00	3,070.00	20,300.00		6.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	196.00	39.20	280.00	0.00
Total actual harvest content	29.38	205.79	45.45	300.54	1,021.55

J&M South

Field name: J&M South

10/29/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 38 Plant date: 10/29/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/01/2023	512.50 *ton*	Dry-weight		66.0	17,600.00	2,390.00	20,200.00		7.93

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	192.00	25.60	217.60	0.00
Total actual harvest content	13.49	161.41	21.92	185.26	727.26

06/02/2023: Corn, silage

Crop: Corn, silage Acres planted: 38 Plant date: 06/02/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/22/2023	1,012.50 *ton*	Dry-weight		48.0	12,400.00	2,920.00	17,800.00		7.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	361.20	84.00	280.00	0.00
Total actual harvest content	26.64	343.61	80.91	493.25	1,939.74

Wagemann East

Field name: Wagemann East

10/29/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 30 Plant date: 10/29/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/08/2023	512.50 *ton*	Dry-weight		64.2	15,500.00	2,130.00	18,500.00		8.48

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	177.60	25.60	211.20	0.00
Total actual harvest content	17.08	189.59	26.05	226.29	1,037.25

06/07/2023: Corn, silage

Crop: Corn, silage Acres planted: 30 Plant date: 06/07/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/26/2023	825.00 *ton*	Dry-weight		77.4	12,300.00	3,390.00	20,100.00		7.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	252.00	0.00
Total actual harvest content	27.50	152.89	42.14	249.84	907.39

Wagemann West

Field name: Wagemann West

10/29/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 40 Plant date: 10/29/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/04/2023	587.50 *ton*	Dry-weight		71.8	22,000.00	2,710.00	25,300.00		9.46

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	198.40	25.60	177.60	0.00
Total actual harvest content	14.69	182.24	22.45	209.58	783.64

Wagemann West

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 40 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/25/2023	1,200.00 *ton*	Dry-weight		68.0	13,000.00	3,280.00	20,200.00		6.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	232.40	58.80	338.80	0.00
Total actual harvest content	30.00	249.60	62.98	387.84	1,248.00

NUTRIENT BUDGET

A. LAND APPLICATIONS

J&M North - 10/29/2022: Oats, silage-soft dough

Field name: J&M North

Crop: Oats, silage-soft dough — Plant date: 10/29/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/03/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MANURE APPLICATION	Scraped material	148.26	54.60	152.04	0.00	60.00 *ton*
Application event totals		148.26	54.60	152.04	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/13/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	70.60	12.03	115.21	1,186.66	360,000.00 *gal*
Application event totals		70.60	12.03	115.21	1,186.66	

J&M North - 06/08/2023: Corn, silage

Field name: J&M North

Crop: Corn, silage — Plant date: 06/08/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
J & M Ag Well	Ground water	5.75	0.00	0.00	0.00	2,808,000.00 *gal*
Application event totals		5.75	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
J & M Ag Well	Ground water	5.03	0.00	0.00	0.00	2,457,000.00 *gal*
Application event totals		5.03	0.00	0.00	0.00	

J&M North - 06/08/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	112.51	33.91	394.30	1,619.26	630,000.00 *gal*
J & M Ag Well	Ground water	3.28	0.00	0.00	0.00	1,599,000.00 *gal*
Application event totals		115.78	33.91	394.30	1,619.26	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
J & M Ag Well	Ground water	4.47	0.00	0.00	0.00	2,184,000.00 *gal*
Application event totals		4.47	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
J & M Ag Well	Ground water	4.47	0.00	0.00	0.00	2,184,000.00 *gal*
Application event totals		4.47	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	56.25	16.95	197.15	809.63	315,000.00 *gal*
J & M Ag Well	Ground water	4.47	0.00	0.00	0.00	2,184,000.00 *gal*
Application event totals		60.73	16.95	197.15	809.63	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
J & M Ag Well	Ground water	4.47	0.00	0.00	0.00	2,184,000.00 *gal*
Application event totals		4.47	0.00	0.00	0.00	

J&M South - 10/29/2022: Oats, silage-soft dough

Field name: J&M South

Crop: Oats, silage-soft dough

Plant date: 10/29/2022

J&M South - 10/29/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/03/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MANURE APPLICATION	Scraped material	130.05	47.89	133.37	0.00	100.00 *ton*
Application event totals		130.05	47.89	133.37	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/14/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	55.74	9.50	90.96	936.84	540,000.00 *gal*
Application event totals		55.74	9.50	90.96	936.84	

J&M South - 06/02/2023: Corn, silage

Field name: J&M South

Crop: Corn, silage

Plant date: 06/02/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
J & M Ag Well	Ground water	5.05	0.00	0.00	0.00	4,680,000.00 *gal*
Application event totals		5.05	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
J & M Ag Well	Ground water	4.67	0.00	0.00	0.00	4,329,000.00 *gal*
Application event totals		4.67	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	114.20	34.42	400.23	1,643.61	1,215,000.00 *gal*
J & M Ag Well	Ground water	2.86	0.00	0.00	0.00	2,652,000.00 *gal*
Application event totals		117.06	34.42	400.23	1,643.61	

J&M South - 06/02/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
J & M Ag Well	Ground water	4.21	0.00	0.00	0.00	3,900,000.00 *gal*
Application event totals		4.21	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
J & M Ag Well	Ground water	4.21	0.00	0.00	0.00	3,900,000.00 *gal*
Application event totals		4.21	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	59.21	17.85	207.53	852.24	630,000.00 *gal*
J & M Ag Well	Ground water	4.21	0.00	0.00	0.00	3,900,000.00 *gal*
Application event totals		63.42	17.85	207.53	852.24	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
J & M Ag Well	Ground water	4.21	0.00	0.00	0.00	3,900,000.00 *gal*
Application event totals		4.21	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
J & M Ag Well	Ground water	4.21	0.00	0.00	0.00	3,900,000.00 *gal*
Application event totals		4.21	0.00	0.00	0.00	

Wagemann East - 10/29/2022: Oats, silage-soft dough

Field name: Wagemann East

Crop: Oats, silage-soft dough

Plant date: 10/29/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Wagemann East - 10/29/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/03/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MANURE APPLICATION	Scraped material	148.26	54.60	152.04	0.00	90.00 *ton*
Application event totals		148.26	54.60	152.04	0.00	

03/01/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	41.18	7.02	67.21	692.22	315,000.00 *gal*
Application event totals		41.18	7.02	67.21	692.22	

Wagemann East - 06/07/2023: Corn, silage

Field name: Wagemann East

Crop: Corn, silage

Plant date: 06/07/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wagemann Large Ag	Ground water	3.17	0.00	0.00	0.00	2,880,000.00 *gal*
Wagemann Small Ag	Ground water	0.33	0.00	0.00	0.00	960,000.00 *gal*
Application event totals		3.50	0.00	0.00	0.00	

06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wagemann Large Ag	Ground water	2.88	0.00	0.00	0.00	2,610,000.00 *gal*
Wagemann Small Ag	Ground water	0.30	0.00	0.00	0.00	870,000.00 *gal*
Application event totals		3.17	0.00	0.00	0.00	

Wagemann East - 06/07/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	75.00	22.61	262.87	1,079.51	630,000.00 *gal*
Wagemann Large Ag	Ground water	2.33	0.00	0.00	0.00	2,115,000.00 *gal*
Wagemann Small Ag	Ground water	0.24	0.00	0.00	0.00	705,000.00 *gal*
Application event totals		77.58	22.61	262.87	1,079.51	

07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wagemann Large Ag	Ground water	2.58	0.00	0.00	0.00	2,340,000.00 *gal*
Wagemann Small Ag	Ground water	0.27	0.00	0.00	0.00	780,000.00 *gal*
Application event totals		2.84	0.00	0.00	0.00	

08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wagemann Large Ag	Ground water	2.58	0.00	0.00	0.00	2,340,000.00 *gal*
Wagemann Small Ag	Ground water	0.27	0.00	0.00	0.00	780,000.00 *gal*
Application event totals		2.84	0.00	0.00	0.00	

08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	75.00	22.61	262.87	1,079.51	630,000.00 *gal*
Wagemann Large Ag	Ground water	2.58	0.00	0.00	0.00	2,340,000.00 *gal*
Wagemann Small Ag	Ground water	0.27	0.00	0.00	0.00	780,000.00 *gal*
Application event totals		77.85	22.61	262.87	1,079.51	

09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wagemann Large Ag	Ground water	2.58	0.00	0.00	0.00	2,340,000.00 *gal*
Wagemann Small Ag	Ground water	0.27	0.00	0.00	0.00	780,000.00 *gal*
Application event totals		2.84	0.00	0.00	0.00	

Wagemann West - 10/29/2022: Oats, silage-soft dough

Field name: Wagemann West

Crop: Oats, silage-soft dough

Plant date: 10/29/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/03/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MANURE APPLICATION	Scraped material	172.97	63.70	177.38	0.00	140.00 *ton*
Application event totals		172.97	63.70	177.38	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/02/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	52.95	9.02	86.41	889.99	540,000.00 *gal*
Application event totals		52.95	9.02	86.41	889.99	

Wagemann West - 06/01/2023: Corn, silage

Field name: Wagemann West

Crop: Corn, silage

Plant date: 06/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wagemann Large Ag	Ground water	3.01	0.00	0.00	0.00	3,645,000.00 *gal*
Wagemann Small Ag	Ground water	0.31	0.00	0.00	0.00	1,215,000.00 *gal*
Application event totals		3.32	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wagemann Large Ag	Ground water	2.75	0.00	0.00	0.00	3,330,000.00 *gal*
Wagemann Small Ag	Ground water	0.28	0.00	0.00	0.00	1,110,000.00 *gal*
Application event totals		3.04	0.00	0.00	0.00	

Wagemann West - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	108.49	32.70	380.22	1,561.43	1,215,000.00 *gal*
Wagemann Large Ag	Ground water	1.82	0.00	0.00	0.00	2,205,000.00 *gal*
Wagemann Small Ag	Ground water	0.19	0.00	0.00	0.00	735,000.00 *gal*
Application event totals		110.50	32.70	380.22	1,561.43	

07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wagemann Large Ag	Ground water	2.57	0.00	0.00	0.00	3,105,000.00 *gal*
Wagemann Small Ag	Ground water	0.27	0.00	0.00	0.00	1,035,000.00 *gal*
Application event totals		2.83	0.00	0.00	0.00	

07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wagemann Large Ag	Ground water	2.57	0.00	0.00	0.00	3,105,000.00 *gal*
Wagemann Small Ag	Ground water	0.27	0.00	0.00	0.00	1,035,000.00 *gal*
Application event totals		2.83	0.00	0.00	0.00	

08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	56.25	16.95	197.15	809.63	630,000.00 *gal*
Wagemann Large Ag	Ground water	2.57	0.00	0.00	0.00	3,105,000.00 *gal*
Wagemann Small Ag	Ground water	0.27	0.00	0.00	0.00	1,035,000.00 *gal*
Application event totals		59.08	16.95	197.15	809.63	

08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wagemann Large Ag	Ground water	2.57	0.00	0.00	0.00	3,105,000.00 *gal*
Wagemann Small Ag	Ground water	0.27	0.00	0.00	0.00	1,035,000.00 *gal*
Application event totals		2.83	0.00	0.00	0.00	

Wagemann West - 06/01/2023: Corn, silage

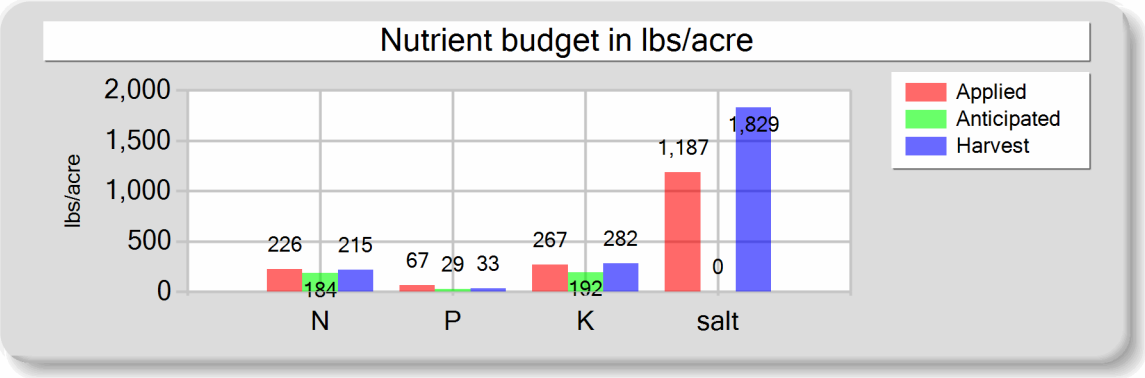
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	28.13	8.48	98.58	404.82	315,000.00 *gal*
Wagemann Large Ag	Ground water	2.57	0.00	0.00	0.00	3,105,000.00 *gal*
Wagemann Small Ag	Ground water	0.27	0.00	0.00	0.00	1,035,000.00 *gal*
Application event totals		30.96	8.48	98.58	404.82	

B. NUTRIENT BUDGET

J&M North - 10/29/2022: Oats, silage-soft dough

Field name: J&M North | Crop: Oats, silage-soft dough | Plant date: 10/29/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	148.26	54.60	152.04	0.00
Process wastewater	70.60	12.03	115.21	1,186.66
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	225.86	66.63	267.25	1,186.66
Anticipated crop nutrient removal	184.00	28.80	192.00	0.00
Actual crop nutrient removal	215.39	33.09	281.56	1,829.48
Nutrient balance	10.47	33.55	-14.31	-642.82
Applied to removed ratio	1.05	2.01	0.95	0.65

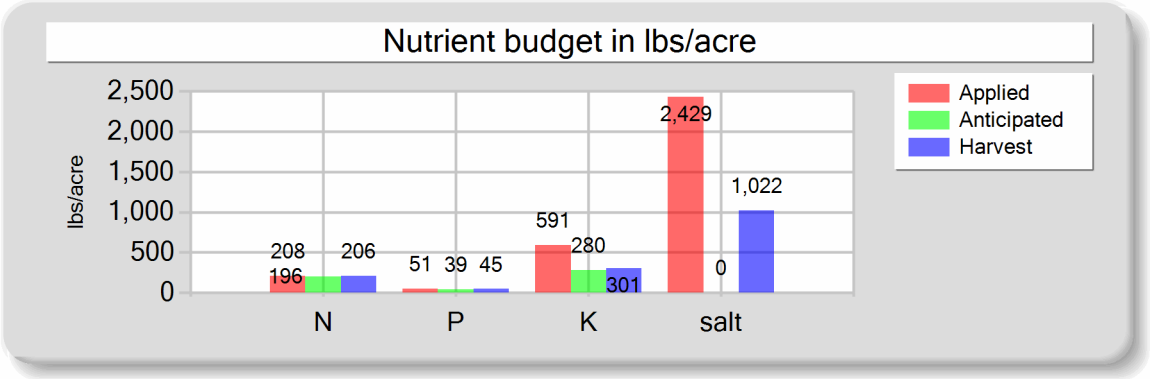
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
360,000.00	*gallons*
13.26	*acre-inches*
0.66	*inches/acre*

Total harvests for the crop	
1	*harvests*

J&M North - 06/08/2023: Corn, silage

Field name: J&M North | Crop: Corn, silage | Plant date: 06/08/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	168.76	50.86	591.45	2,428.90
Fresh water	31.96	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	207.72	50.86	591.45	2,428.90
Anticipated crop nutrient removal	196.00	39.20	280.00	0.00
Actual crop nutrient removal	205.79	45.45	300.54	1,021.55
Nutrient balance	1.93	5.41	290.91	1,407.35
Applied to removed ratio	1.01	1.12	1.97	2.38

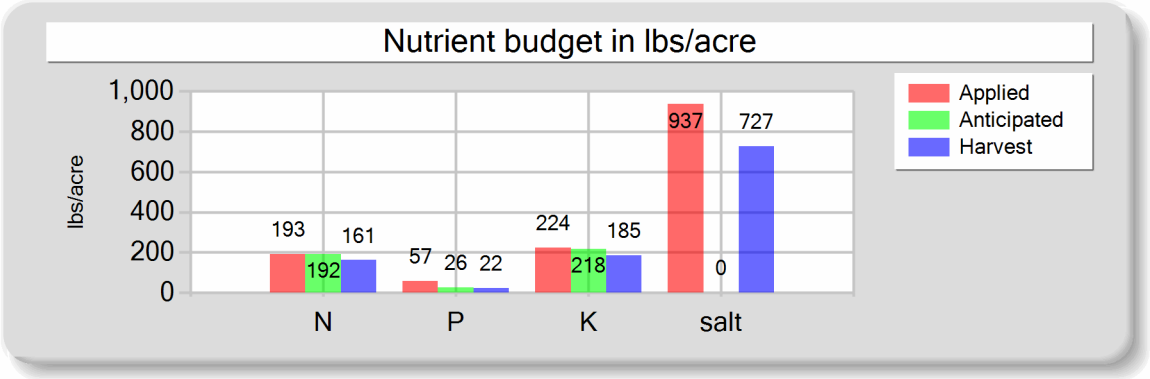
Fresh water applied	
15,600,000.00	*gallons*
574.49	*acre-inches*
28.72	*inches/acre*

Process wastewater applied	
945,000.00	*gallons*
34.80	*acre-inches*
1.74	*inches/acre*

Total harvests for the crop	
1	*harvests*

J&M South - 10/29/2022: Oats, silage-soft dough

Field name: J&M South | Crop: Oats, silage-soft dough | Plant date: 10/29/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	130.05	47.89	133.37	0.00
Process wastewater	55.74	9.50	90.96	936.84
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	192.79	57.39	224.32	936.84
Anticipated crop nutrient removal	192.00	25.60	217.60	0.00
Actual crop nutrient removal	161.41	21.92	185.26	727.26
Nutrient balance	31.38	35.47	39.07	209.57
Applied to removed ratio	1.19	2.62	1.21	1.29

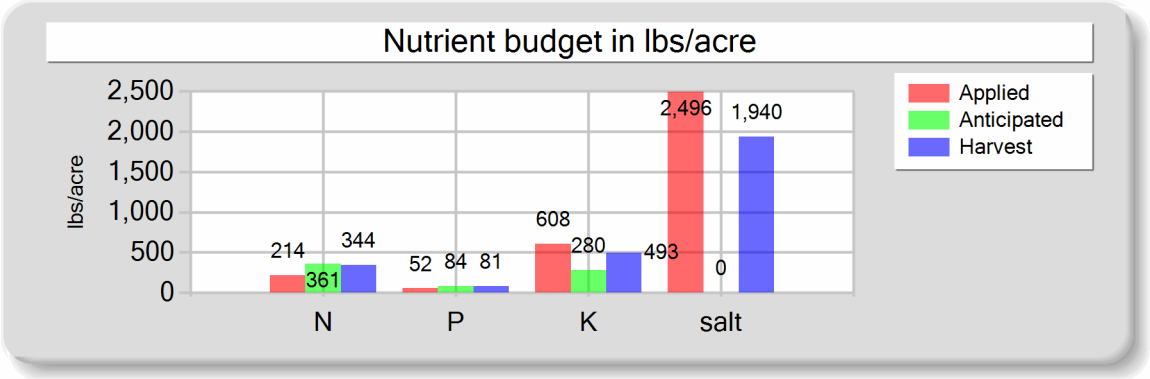
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
540,000.00 *gallons*
19.89 *acre-inches*
0.52 *inches/acre*

Total harvests for the crop
1 *harvests*

J&M South - 06/02/2023: Corn, silage

Field name: J&M South | Crop: Corn, silage | Plant date: 06/02/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	173.41	52.27	607.76	2,495.86
Fresh water	33.60	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	214.01	52.27	607.76	2,495.86
Anticipated crop nutrient removal	361.20	84.00	280.00	0.00
Actual crop nutrient removal	343.61	80.91	493.25	1,939.74
Nutrient balance	-129.60	-28.65	114.51	556.12
Applied to removed ratio	0.62	0.65	1.23	1.29

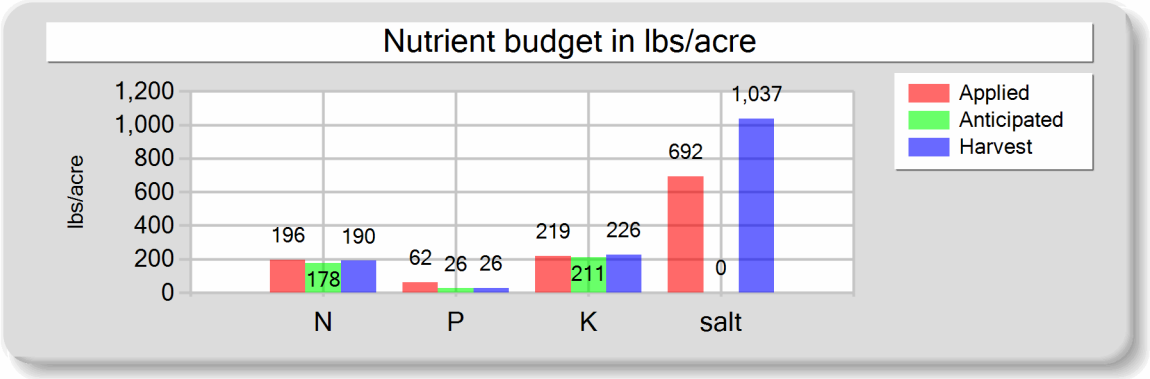
Fresh water applied	
31,161,000.00	*gallons*
1,147.55	*acre-inches*
30.20	*inches/acre*

Process wastewater applied	
1,845,000.00	*gallons*
67.95	*acre-inches*
1.79	*inches/acre*

Total harvests for the crop	
1	*harvests*

Wagemann East - 10/29/2022: Oats, silage-soft dough

Field name: Wagemann East | Crop: Oats, silage-soft dough | Plant date: 10/29/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	148.26	54.60	152.04	0.00
Process wastewater	41.18	7.02	67.21	692.22
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	196.44	61.62	219.25	692.22
Anticipated crop nutrient removal	177.60	25.60	211.20	0.00
Actual crop nutrient removal	189.59	26.05	226.29	1,037.25
Nutrient balance	6.85	35.57	-7.04	-345.03
Applied to removed ratio	1.04	2.37	0.97	0.67

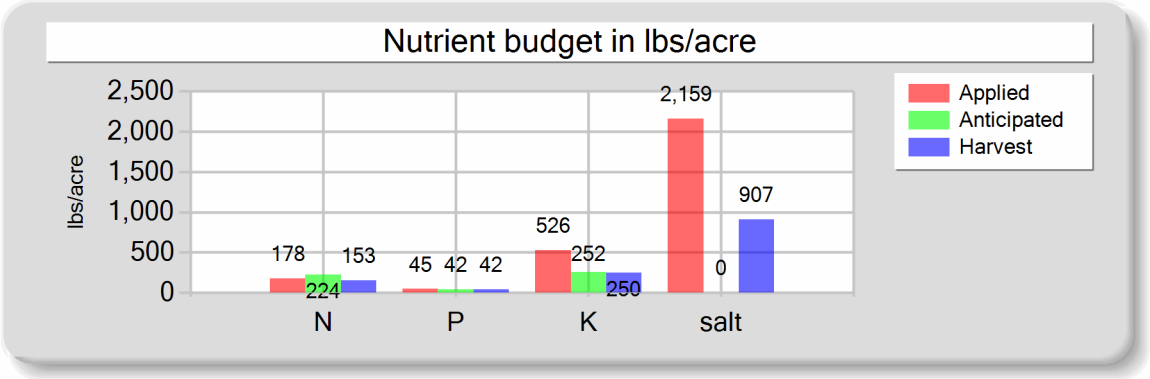
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
315,000.00 *gallons*
11.60 *acre-inches*
0.39 *inches/acre*

Total harvests for the crop
1 *harvests*

Wagemann East - 06/07/2023: Corn, silage

Field name: Wagemann East | Crop: Corn, silage | Plant date: 06/07/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	150.01	45.21	525.74	2,159.02
Fresh water	20.62	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	177.63	45.21	525.74	2,159.02
Anticipated crop nutrient removal	224.00	42.00	252.00	0.00
Actual crop nutrient removal	152.89	42.14	249.84	907.39
Nutrient balance	24.74	3.08	275.89	1,251.63
Applied to removed ratio	1.16	1.07	2.10	2.38

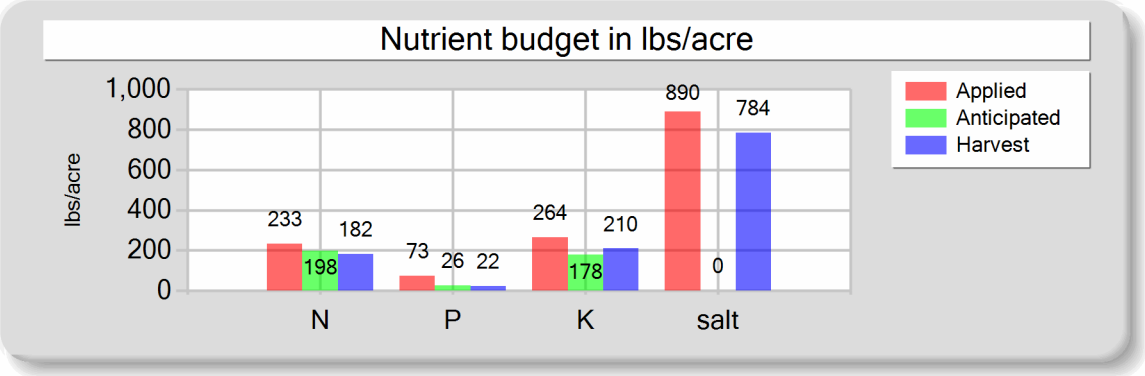
Fresh water applied	
22,620,000.00	*gallons*
833.02	*acre-inches*
27.77	*inches/acre*

Process wastewater applied	
1,260,000.00	*gallons*
46.40	*acre-inches*
1.55	*inches/acre*

Total harvests for the crop	
1	*harvests*

Wagemann West - 10/29/2022: Oats, silage-soft dough

Field name: Wagemann West | Crop: Oats, silage-soft dough | Plant date: 10/29/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	172.97	63.70	177.38	0.00
Process wastewater	52.95	9.02	86.41	889.99
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	232.92	72.72	263.79	889.99
Anticipated crop nutrient removal	198.40	25.60	177.60	0.00
Actual crop nutrient removal	182.24	22.45	209.58	783.64
Nutrient balance	50.68	50.27	54.21	106.35
Applied to removed ratio	1.28	3.24	1.26	1.14

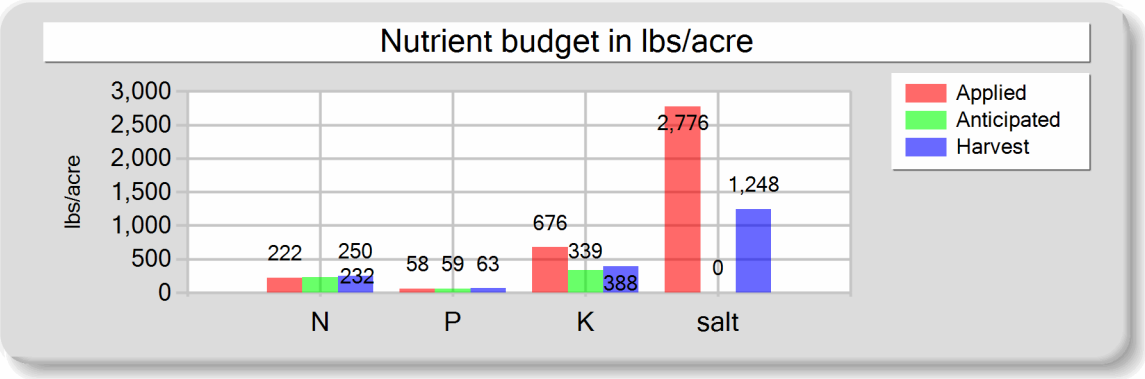
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
540,000.00 *gallons*
19.89 *acre-inches*
0.50 *inches/acre*

Total harvests for the crop
1 *harvests*

Wagemann West - 06/01/2023: Corn, silage

Field name: Wagemann West Crop: Corn, silage Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	192.87	58.13	675.95	2,775.88
Fresh water	22.52	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	222.39	58.13	675.95	2,775.88
Anticipated crop nutrient removal	232.40	58.80	338.80	0.00
Actual crop nutrient removal	249.60	62.98	387.84	1,248.00
Nutrient balance	-27.21	-4.84	288.11	1,527.88
Applied to removed ratio	0.89	0.92	1.74	2.22

Fresh water applied	
32,940,000.00	*gallons*
1,213.07	*acre-inches*
30.33	*inches/acre*

Process wastewater applied	
2,160,000.00	*gallons*
79.55	*acre-inches*
1.99	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

T2301026B

Sample and source description: T2301026B

Sample date: 10/28/2022 Material type: Scraped material Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 30.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	35,300.00	13,000.00	36,200.00	0.00	0.00	0.00	0.00	0.00		0.00
DL	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00		1.00

T3130027F

Sample and source description: T3130027F

Sample date: 05/10/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 22.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,300.00	5,400.00	9,010.00							0.00
DL	1.00	1.00	1.00							1.00

T3264043B

Sample and source description: T3264043B

Sample date: 09/21/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 31.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,200.00	3,830.00	15,400.00	18,600.00	8,110.00	4,240.00	5,680.00	1,310.00		70.70
DL	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00		1.00

T3335008A

Sample and source description: T3335008A

Sample date: 12/01/2023 Material type: Scraped material Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 17.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	25,600.00	6,890.00	13,500.00							0.00
DL	1.00	1.00	1.00							1.00

B. PROCESS WASTEWATER ANALYSES

T3059012A

Sample and source description: T3059012A

Sample date: 02/28/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	470.00	297.00	0.00	0.00	80.10	767.00								5,990.00	7,900
DL	1.00	1.00	1.00	1.00	1.00	1.00								100.00	1

T3130027E

Sample and source description: T3130027E

Sample date: 05/10/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	596.00	384.00	0.00	0.00	68.40	881.00								7,690.00	5,220
DL	1.00	1.00	1.00	1.00	1.00	1.00								100.00	1

T3264043A

Sample and source description: T3264043A

Sample date: 09/21/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.80

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	428.00	339.00	0.00	0.00	129.00	1,500.00	400.00	238.00	400.00	3,100.00	0.00	17.10	393.00	6,710.00	6,160
DL	1.00	1.00	1.00	0.14	1.00	1.00	1.00	1.00	1.00	1.00	2.00	1.00	1.00	100.00	1

C. FRESH WATER ANALYSES

J & M Ag Well

T3314015D

Sample description: T3314015D

Sample date: 11/10/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	4.91	0.00	4.91								734.00	0
DL	1.00	0.50	1.00								100.00	1

J & M Dom. #1

T3314015E

Sample description: T3314015E

Sample date: 11/10/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	2.64	0.00	2.64								902.00	0
DL	1.00	0.50	1.00								100.00	1

J & M Dom. #2

J & M Dom. #2

T3314015F

Sample description: T3314015F

Sample date: 11/10/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								855.00	0
DL	0.14	0.50	0.14								100.00	1

Wagemann Dom.

T3314015A

Sample description: T3314015A

Sample date: 11/10/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	5.83	0.00	5.83	58.40	35.80	31.80	242.00	0.00	38.00	53.40	708.00	350
DL	1.00	0.50	1.00	1.00	1.00	1.00	1.00	2.00	1.00	1.00	100.00	1

Wagemann Large Ag

T3314015C

Sample description: T3314015C

Sample date: 11/10/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	3.96	0.00	3.96								785.00	0
DL	1.00	0.50	1.00								100.00	1

Wagemann Small Ag

Wagemann Small Ag

T3314015B

Sample description: T3314015B

Sample date: 11/10/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.23	0.00	1.23								980.00	0
DL	1.00	0.50	1.00								100.00	1

D. SOIL ANALYSES

J&M North

T3349017A

Sample and source description: T3349017A

Sample date: 12/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			36.00		0.00	0.00	
DL			1.00		1.00	1.00	

J&M South

T3349017B

Sample and source description: T3349017B

Sample date: 12/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			46.00		0.00	0.00	
DL			1.00		1.00	1.00	

Wagemann East

Wagemann East

T3349017C

Sample and source description: T3349017C

Sample date: 12/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			46.00		0.00	0.00	
DL			1.00		1.00	1.00	

Wagemann West

T3349017D

Sample and source description: T3349017D

Sample date: 12/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			60.00		0.00	0.00	
DL			1.00		1.00	1.00	

E. PLANT TISSUE ANALYSES

J&M North - 10/29/2022: Oats, silage-soft dough

T3130027A

Sample and source description: T3130027A

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,600.00	2,550.00	21,700.00		14.10
DL	1.00	1.00	1.00		1.00

J&M North - 06/08/2023: Corn, silage

J&M North - 06/08/2023: Corn, silage

T3272038D

Sample and source description: T3272038D

Sample date: 09/29/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 74.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,900.00	3,070.00	20,300.00		6.90
DL	1.00	1.00	1.00		1.00

J&M South - 10/29/2022: Oats, silage-soft dough

T3130027B

Sample and source description: T3130027B

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,600.00	2,390.00	20,200.00		7.93
DL	1.00	1.00	1.00		1.00

J&M South - 06/02/2023: Corn, silage

T3272038A

Sample and source description: T3272038A

Sample date: 09/29/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 48.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,400.00	2,920.00	17,800.00		7.00
DL	1.00	1.00	1.00		1.00

Wagemann East - 10/29/2022: Oats, silage-soft dough

T3130027C

Sample and source description: T3130027C

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,500.00	2,130.00	18,500.00		8.48
DL	1.00	1.00	1.00		1.00

Wagemann East - 06/07/2023: Corn, silage

T3272038B

Sample and source description: T3272038B

Sample date: 09/29/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 77.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,300.00	3,390.00	20,100.00		7.30
DL	1.00	1.00	1.00		1.00

Wagemann West - 10/29/2022: Oats, silage-soft dough

T3130027D

Sample and source description: T3130027D

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,000.00	2,710.00	25,300.00		9.46
DL	1.00	1.00	1.00		1.00

Wagemann West - 06/01/2023: Corn, silage

T3272038C

Sample and source description: T3272038C

Sample date: 09/29/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,000.00	3,280.00	20,200.00		6.50
DL	1.00	1.00	1.00		1.00

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

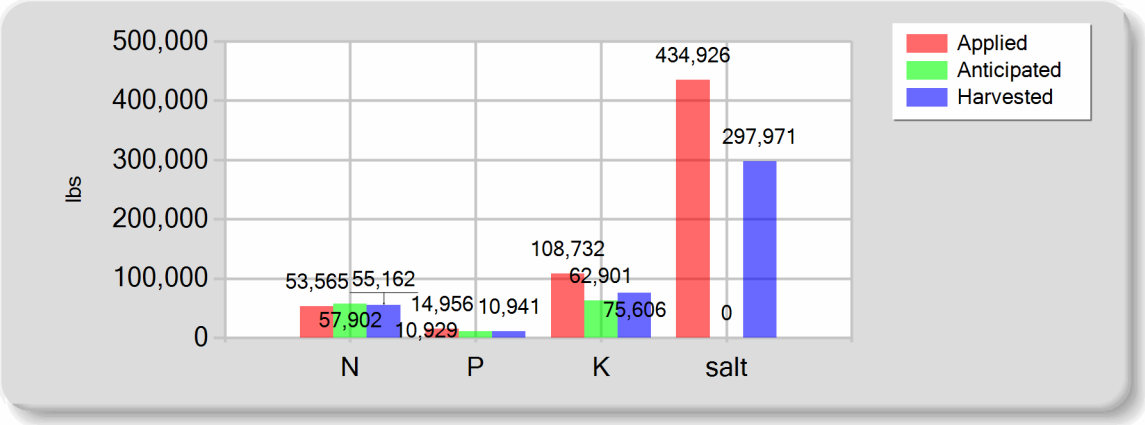
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

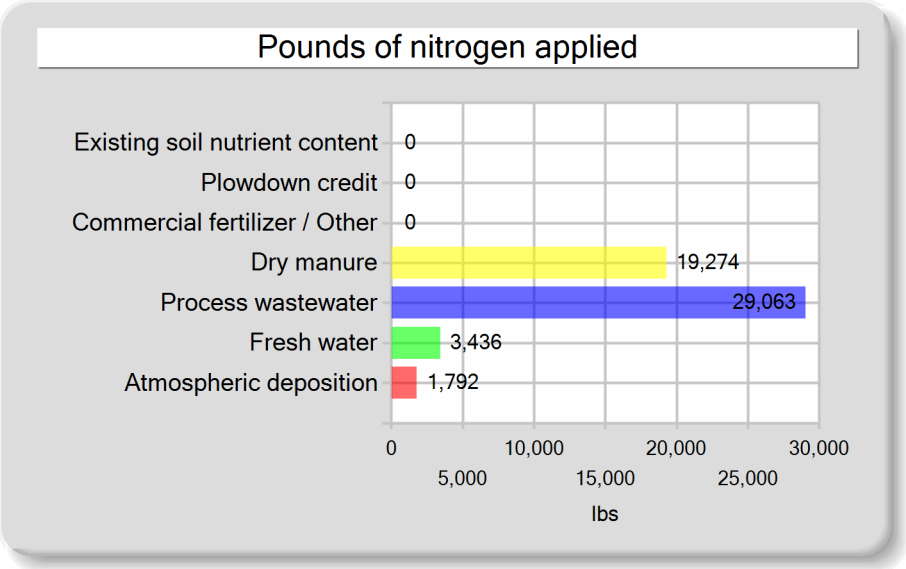
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

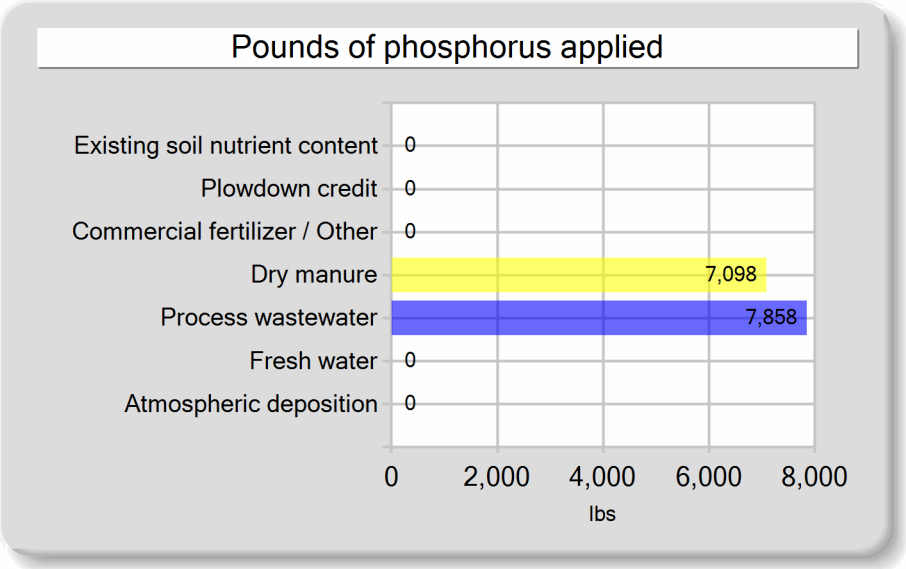
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	19,273.80	7,098.00	19,765.20	0.00
Process wastewater	29,063.38	7,858.20	88,966.75	434,925.54
Fresh water	3,435.59	0.00	0.00	0.00
Atmospheric deposition	1,792.00	0.00	0.00	0.00
Total nutrients applied	53,564.77	14,956.20	108,731.95	434,925.54
Anticipated crop nutrient removal	57,901.60	10,928.80	62,900.80	0.00
Actual crop nutrient removal	55,162.39	10,941.16	75,605.70	297,971.22
Nutrient balance	-1,597.61	4,015.04	33,126.25	136,954.32
Applied to removed ratio	0.97	1.37	1.44	1.46

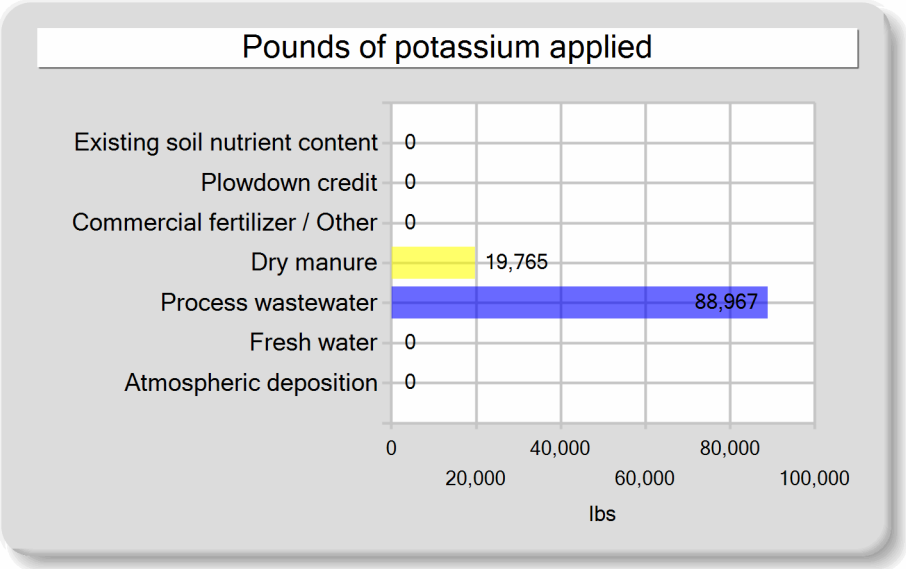
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

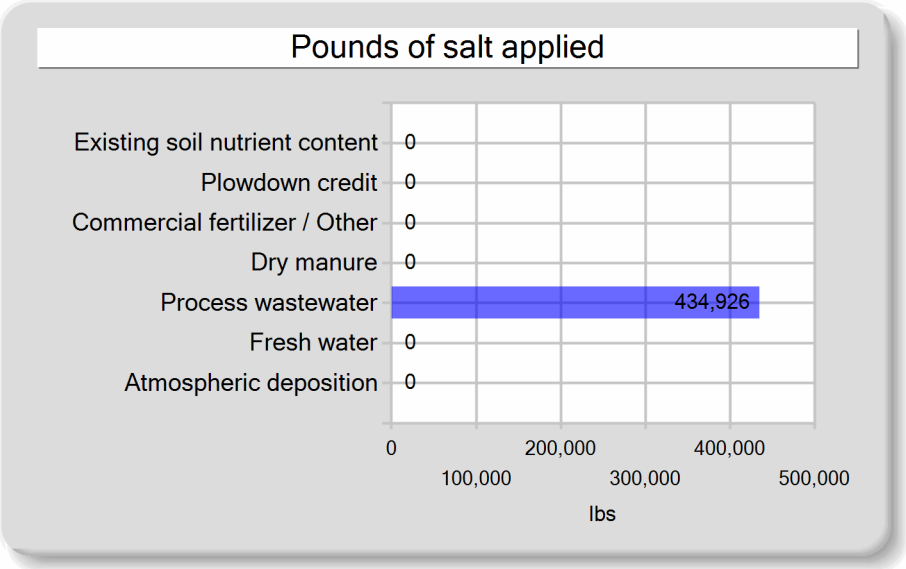


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

Annual Report - General Order No. R5-2007-0035

Reporting period 01/01/2023 to 12/31/2023.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

John F Duarte	
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
John Fernando Duarte	Todd Anthony Duarte
PRINT OR TYPE NAME	PRINT OR TYPE NAME
6/27/24	6/27/24
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination**.
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: ___TODD DUARTE_____

Name of Dairy Facility: _J & M DAIRY_____

Facility Address: __11910 BRUCEVILLE RD._____ELK GROVE_____95757__
Number and Street City Zip Code

Contact Person Name and Phone Number:_TODD DUARTE _____(916) 952-8355__
Name Phone Number

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: __M. MARAVILLA TRUCKING INC._____

Address of Hauling Company /Person:_175 ASH AVE._____MENDOTA_____93640___
Number and Street City Zip Code

Contact Person:____MARIO MARAVILLA_____(559) 698-4312_____
Name Phone Number

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) _____ (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

_RIVERBEND VINEYARDS__24521 MC INTIRE RD.CLEMENTS 95227__(559) 698-4312____
Name Number and Street City Zip Code Phone Number

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

_____SAME_____
Number and Street City Zip Code Assessor's Parcel Number

Dates Hauled: __515 – 10/31/23_____

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount: N 1.42 % P 0.383% K 1.54 %

Manure: _150_____Tons or Cubic Yards (indicate which units used)
Manure Solids Content (if amount reported in tons): _68.40%_____
Manure Density (if amount reported in cubic yards): _____

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount

Process Wastewater _____ Gallons

Method used to determine volume of process wastewater _____

_____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

____ Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after 31 December 2007 to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: _____ Date: 6-27-24

Hauler's Signature: _____ Date: 6-27-24

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination**.
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: ___TODD DUARTE_____

Name of Dairy Facility: _J & M DAIRY_____

Facility Address: __11910 BRUCEVILLE RD._____ELK GROVE_____95757__
Number and Street City Zip Code

Contact Person Name and Phone Number:_TODD DUARTE _____(916) 952-8355__
Name Phone Number

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: __M. MARAVILLA TRUCKING INC._____

Address of Hauling Company /Person:_175 ASH AVE._____MENDOTA_____93640___
Number and Street City Zip Code

Contact Person:____MARIO MARAVILLA_____(559) 698-4312_____
Name Phone Number

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) _____ (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

_RIVERBEND VINEYARDS__15645 N. HWY 88 LOCKEFORD 95237__(559) 698-4312____
Name Number and Street City Zip Code Phone Number

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

_____SAME_____
Number and Street City Zip Code Assessor's Parcel Number

Dates Hauled: __515 – 10/31/23_____

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount: N 1.83 % P 0.540% K 0.901 %

Manure: _350_____Tons or Cubic Yards (indicate which units used)
Manure Solids Content (if amount reported in tons): _77.80%_____
Manure Density (if amount reported in cubic yards): _____

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount

Process Wastewater _____ Gallons

Method used to determine volume of process wastewater _____

_____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

____ Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after 31 December 2007 to such party _____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: _____ Date: 6-27-24

Hauler's Signature: _____ Date: 6-27-24

DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380

PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Authorized Copy Released To:
JCRK, INC.
(209)327-0992

Client: J&M DAIRY
Mailing address: 11910 BRUCEVILLE RD.
(not facility address) ELK GROVE, CA 95757
Phone#: (916)952-8355

Sampled By: Todd Duarte
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
H1 EC*, NO3-N, NH4-N*
H6 EC, TN, TDS
H7 Combination of H1 & H6
H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
H3 EC*, NH4-N, TKN, TP, K, TDS
H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
H8 Combination of H3 &H5
H9 Combination of H4 &H5

TILE DRAIN
H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*
___ (Check here if performed)

DISCHARGE WATER
E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
M1 % Moisture
M3 TN TP, K, % Moisture
M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
M7 Combination of M3 & M4

PLANT TISSUE
F18 % Moisture, Ash, TN, P, K, DM and as received
F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
K3 Soluble Phosphorus
K4 1ft NO3, Organic Matter
K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
K7 Combination of K3 & K4
K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	11/8/23	8:30a	Wagemann Domestic	Dom. Well	11.0°C	20 min					H14
2	11/9/23	8:20a	Wagemann Small Ag	Ag Well	10.8'	✓	X				
3	11/8/23	9:15a	Wagemann Large Ag	✓	10.8'	✓	X				
4	11/9/23	10:05a	J+M Ag Well	✓	10.6'	✓	X				
5	✓	8:45a	J+M Dom. #1	Dom. Well	9.8'	✓	X				
6	✓	9:20a	J+M Dom. #2	✓	10.0'	✓	X				
7											
8											
9											
10											

Analysis

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	(LABORATORY USE ONLY) Logged In By:	LABORATORY NUMBER
1st	Jacquie Captein	JCRK		11/10/23 8:00a		
2nd		DENELE ANALYTICAL, INC.	11-10-23 1:00			T3314015

Required to be submitted to any agency? (YES / NO)

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
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Compliance Analysis Report

Customer
J & M Dairy - Elk Grove
11910 Bruceville Rd.
Elk Grove, CA 95757

Grower

Consultant:
Jacquie Captein
P.O. Box 386
Herald, CA 95638

PURCHASE ORDER:
RECEIVED DATE: 11/10/23 1:00 PM
SUBMITTED BY: Jacquie
ANALYZED DATE: 11/13/23

Irrigation /Ground Water (H14)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 11.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Wagemann Dom.	T3314015A	Electrical Conductivity (EC)	EPA 120.1 and	0.708	mmhos/cm
		Bicarbonate Alkalinity (ALK-B)	SM 2320B-2011	242	mg/L
		Carbonate Alkalinity (Carb)	SM 2320B-2011	< 2.00	mg/L
		Sulfate (SO4)	EPA 375.2	38.0	mg/L
		Sodium (Na)	EPA 200.7	31.8	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Magnesium (Mg)	EPA 200.7	35.8	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	5.83	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	350	mg/L
		Calcium (Ca)	EPA 200.7	58.4	mg/L
		Chloride (Cl)	SM 4500-Cl D-2011	53.4	mg/L
		Soluble Salts (SALT-SOL)	Calculation	453	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	94.4			0.000 (T3314013A)	0.000 (null)	
Calcium (Ca)	105	94.5 (T3314002A)	94.4 (T3314002A)			0.102
Chloride (Cl)	101	91.1 (T3314001A)	90.2 (T3314001A)			0.927
Magnesium (Mg)	94.1	103 (T3314002A)	102 (T3314002A)			1.14
Nitrate Nitrogen (NO3-N)	100	106 (T3314001A)	103 (T3314001A)			3.44
Sodium (Na)	93.2	110 (T3314002A)	107 (T3314002A)			2.17
Sulfate (SO4)	108	126 (T3314001A)	126 (T3314001A)			0.615
Total Dissolved Solids (TDS)	102			50.0 (T3313017B)	50.0 (Dup)	0.000

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ELAP Certificate No.: 2714

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Compliance Analysis Report

Customer
J & M Dairy - Elk Grove
11910 Bruceville Rd.
Elk Grove, CA 95757

Grower

Consultant:
Jacquie Captein
P.O. Box 386
Herald, CA 95638

PURCHASE ORDER:
RECEIVED DATE: 11/10/23 1:00 PM
SUBMITTED BY: Jacquie
ANALYZED DATE: 11/13/23

Irrigation/ Ground Water (H1)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 10.8c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Wagemann Small Ag.	T3314015B	Electrical Conductivity (EC)	EPA 120.1 and	0.980	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	1.23	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	627	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	94.4			
Electrical Conductivity (EC)	93.8			1.53
Nitrate Nitrogen (NO3-N)	100	106 (T3314001A)	103 (T3314001A)	3.44

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Grower

Consultant:
Jacquie Captein
P.O. Box 386
Herald, CA 95638

PURCHASE ORDER:
RECEIVED DATE: 11/10/23 1:00 PM
SUBMITTED BY: Jacquie
ANALYZED DATE: 11/13/23

Irrigation/ Ground Water (H1)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 10.8c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Wagemann Large Ag.	T3314015C	Electrical Conductivity (EC)	EPA 120.1 and	0.785	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	3.96	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	502	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	94.4			
Electrical Conductivity (EC)	93.8			1.53
Nitrate Nitrogen (NO3-N)	100	106 (T3314001A)	103 (T3314001A)	3.44

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Grower

Consultant:
Jacquie Captein
P.O. Box 386
Herald, CA 95638

PURCHASE ORDER:
RECEIVED DATE: 11/10/23 1:00 PM
SUBMITTED BY: Jacquie
ANALYZED DATE: 11/14/23

Irrigation/ Ground Water (H1)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 10.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
J & M Ag Well	T3314015D	Electrical Conductivity (EC)	EPA 120.1 and	0.734	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	4.91	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	470	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	94.4			
Electrical Conductivity (EC)	93.8			1.53
Nitrate Nitrogen (NO3-N)	102	100 (T3314015D)	100.0 (T3314015D)	0.459

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Grower

Consultant:
Jacquie Captein
P.O. Box 386
Herald, CA 95638

PURCHASE ORDER:
RECEIVED DATE: 11/10/23 1:00 PM
SUBMITTED BY: Jacquie
ANALYZED DATE: 11/14/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 9.8c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
J & M Dom. #1	T3314015E	Electrical Conductivity (EC)	EPA 120.1 and	0.902	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	2.64	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	578	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	94.4			
Electrical Conductivity (EC)	93.8			1.53
Nitrate Nitrogen (NO3-N)	102	100 (T3314015D)	100.0 (T3314015D)	0.459

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RECEIVED DATE: 11/10/23 1:00 PM
SUBMITTED BY: Jacquie
ANALYZED DATE: 11/14/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 10.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
J & M Dom. #2	T3314015F	Electrical Conductivity (EC)	EPA 120.1 and	0.855	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	< 0.140	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	547	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	94.4			
Electrical Conductivity (EC)	93.8			1.53
Nitrate Nitrogen (NO3-N)	102	100 (T3314015D)	100.0 (T3314015D)	0.459

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